Yuhang *Zhang*

Research Summary — TexSyn VII Submission, May 2026

Houston, TX · +1 (979) 327-9652 · yuhangzhang36@gmail.com ·
linkedin.com/in/yuhang-zhang-21366a210 · schlenk36.github.io

OVERVIEW

I am a Ph.D. candidate in chemistry at the University of Houston (expected 2027) working in the May group on **complex molecule synthesis** and **asymmetric catalysis**. My primary project is the synthesis and complex assembly of a 2nd-generation chiral bidentate dirhodium(II) biscarboxylate catalyst for enantioselective carbene cascade reactions, where I serve as the lead contributor following the initial design by Dr. H. Barzegar. In parallel, I have pursued **medicinal chemistry** side-projects (warfarin derivatives and flavone scaffolds) and earlier **polymer chemistry** work toward fluorinated anion exchange membrane candidates. I hold a B.S. in Chemical Engineering (Technion, China campus), which complements my synthetic training with process and plant-design foundations.

COMPLEX MOLECULE SYNTHESIS & ASYMMETRIC CATALYSIS

My lead project targets a chiral bidentate Rh_2(II) biscarboxylate catalyst featuring **α-quaternary stereocenters**, a fused-cyclohexane architecture, and an m-xylene tether that bridges both Rh centers at ~90° bite angle to enable enantioselective carbene cascade reactions. The 2nd-generation framework was initially designed by Dr. H. Barzegar; I took over as the lead contributor and executed the **7-step total synthesis** of the bidentate ligand. Key intermediates have been carried through at hundreds-of-milligrams to gram scale. A major optimization milestone was scaling the $PdCl_2$/PMHS ketone reduction from 100 mg to 750 mg at ~100% isolated yield by tuning reagent dispersion and a 23–40 °C water-bath thermal profile across >10 condition screens.

The full ligand has been synthesized; I am currently assembling the metal complex. I presented this work as a poster at TexSyn 2026 (*Y. Zhang, H. Barzegar, J. A. May*). I also support computational analysis of the Rh-carbene pocket in a supporting role.

CHEMICAL METHOD DEVELOPMENT

Beyond catalyst construction, I have developed practical air-/moisture-free workflows for sensitive organometallic chemistry: Schlenk-line technique, freeze-pump-thaw degassing (reproduced at gram scale across multiple runs), DIPA distillation over CaH_2, low-temperature (−78 °C) Li-amide chemistry, and inert-atmosphere reagent transfer. These methods underpin both the dirhodium ligand synthesis and related Pd-mediated reductions.

For my Oral Research Proposal, I synthesized warfarin derivatives using **R-TN1-organocatalyzed conjugate addition**, demonstrating enantioselective organocatalysis on a medicinally relevant scaffold. I have also carried out extensive chiral HPLC method screening for ee determination, evaluating multiple column and solvent systems.

POLYMER CHEMISTRY

From January to September 2024, I worked jointly in the May and Ren groups at the Texas Center for Superconductivity on fluorinated **poly(arylene piperidinium) hydroxide** candidates aimed at anion exchange membrane (AEM) applications for water electrolysis. I synthesized novel polymer candidates and investigated structure–viscosity relationships, with polymer structure and end-group identity characterized by NMR and MALDI-TOF. I presented this work jointly with warfarin-derivative organocatalysis at TexSyn 2024 (*Y. Zhang, C. Yuan, Z. Ren, J. A. May*).

MEDICINAL CHEMISTRY INTEREST

I am actively developing a flavone scaffold program as a medicinal-chemistry extension of my graduate work. I have synthesized the parent flavone scaffold at gram scale (~1.4 g per batch over four batches) and designed an arylboronic-acid-driven analog series for future biological-activity studies. This work aligns with my interest in applying synthetic organic chemistry to drug-discovery-relevant scaffolds, and I am eager to expand this direction in an industrial medicinal chemistry setting.

TECHNICAL PROFICIENCIES

Synthesis: multi-step total synthesis, asymmetric catalyst synthesis & optimization (Rh_2(II) carboxylate), organometallic complex assembly, organocatalysis (R-TN1), reaction condition screening and gram-scale scale-up, polymer synthesis and characterization.

Characterization: multinuclear NMR (^{1}H, ^{13}C, COSY, HSQC, HMBC), chiral HPLC, GC-MS, MALDI-TOF, IR, X-ray crystallography, TLC, recrystallization, TGA, DSC.

Air-free & data: Schlenk line, freeze-pump-thaw, cannula transfer, solvent/amine distillation, ChemDraw, Python/Pandas/Jupyter, with machine-learning and data-science coursework certified through the Hewlett Packard Enterprise Data Science Institute at the University of Houston (*Selected Topics in AI, Intro to Deep Learning, Intro to Machine Learning, Principles of Data Management*).

OUTLOOK

I am seeking roles in **medicinal chemistry, synthesis chemist / R&D, and process development** at U.S. pharma and biotech organizations. My immediate goal is to complete validation of the dirhodium catalyst and advance it toward enantioselective carbene cascade applications. While my training is strongest in complex molecule synthesis, asymmetric catalysis, and polymer chemistry, I am motivated to grow into chemical-biology-adjacent areas through industrial collaboration. I welcome opportunities to discuss how my synthetic and analytical skill set can contribute to drug discovery and method-development teams.